Fiscal Year:	FY 2011	**Task Last Updated:**	FY 04/22/2011
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	RNA Binding Proteins as Evolutionarily Conserved Cellular Spaceflight Response Mechanisms		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Cell Culture		
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City:	Tempe	**State:**	AZ
Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	05/01/2009	**End Date:**	04/30/2012
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Cohen, Jacob	**Contact Phone:**	650-604-3621
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-131		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ott, C. Mark (NASA Johnson Space Center) Wilson, James (Villanova University) Alverdy, John (University of Chicago School of Medicine)		
Grant/Contract No.:	NNX09AH40G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EDITOR'S NOTE: Ground and Flight portions combined into one project per Ames Research Center (July 2010) Understanding infectious disease risks during spaceflight is critical to provide safe passage for human exploration to the moon and Mars. This issue is especially important given reports that the crew's immune system is dysfunctional during flight, and our recent discovery that spaceflight increased the disease-causing potential and globally altered the gene expression of the human food-borne pathogen Salmonella typhimurium. Our research also showed that many of the observed changes in Salmonella were associated with specific proteins in the cells, called RNA binding proteins, that bind to and regulate genetic material within the cell to control cellular characteristics in response to stress. In addition, our studies during spaceflight and using ground-based, spaceflight analogue culture systems have shown that increasing the concentration of certain ionic salts, like phosphates, in the growth media can prevent the increased disease causing potential observed for Salmonella during spaceflight. Interestingly, human cells also contain evolutionarily conserved RNA binding proteins that are very similar in structure and function to those found in bacteria. In human cells, these RNA binding proteins are critical for the cells proper function, including supporting a healthy immune system, maintaining tissue differentiation, and responding to cellular stress. We hypothesize that RNA binding regulatory proteins and their small RNA binding counterparts are key to a conserved, common cellular spaceflight response mechanism in both bacterial and human cells and that this response can be manipulated by environmental salt/ion levels. The implications of a universally conserved molecular response to spaceflight would affect NASA's approach to infectious disease risk assessment, development of biological processing systems for exploration, and changes in biodegradation during a mission. Indeed, if these responses are also observed in human cells, they may translate into knowledge of both the astronaut's response to infection and possibly larger physiological responses that are influenced by cellular changes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will utilize NASA ground-based biotechnology 1) to broaden our knowledge of the host-pathogen interaction between mammalian and microbial cells that leads to infectious disease, and 2) for the development of new therapeutic strategies to combat infectious disease for the general public.
Task Progress:	Previously published results from our lab showed that Salmonella grown in rich LB media in spaceflight and spaceflight analogue conditions (low shear modeled microgravity/LSMMG) displayed increased virulence and global alterations in global gene expression. A central role was identified for the evolutionarily conserved, small regulatory RNA-binding protein Hfq in regulating key aspects of the Salmonella spaceflight and LSMMG response. Interestingly, Salmonella grown in minimal M9 media during spaceflight did not exhibit an increase in virulence, but did demonstrate differential expression of many genes, including those associated with either the regulation of, or regulation by, the Hfq protein and small regulatory RNAs. Moreover, we found that environmental ion concentrations mitigated the enhanced virulence observed in spaceflight-grown Salmonella, and also altered its LSMMG-induced acid stress pathogenesis related response. This led us to hypothesize that RNA binding regulatory proteins and their small RNA binding counterparts are key to a common conserved cellular spaceflight and LSMMG response mechanism. In support of this hypothesis, a recently published joint paper between our lab and that of our collaborators has shown the involvement of Hfq in regulating the LSMMG response of the opportunistic pathogen Pseudomonas aeruginosa (Crabbe et al, 2010, Environ Microbiol). The gene coding for Hfq and several genes belonging to the Hfq regulon were differentially regulated when the transcriptional profiles of P. aeruginosa PAO1 grown in LSMMG and the normal orientation control were compared. Specifically, this study showed for the first time that cultivation of P. aeruginosa PAO1 in the LSMMG environment of the RWV induces a transcriptomic and physiologic response with possible implications for virulence. AlgU was identified as one of the key regulators in the LSMMG-induced response and this alternative sigma factor presumably up-regulated the transcription of genes downstream involved in environmental stress resistance, alginate production and microaerophilic/anaerobic metabolism. Moreover, AlgU might control hfq transcription in P. aeruginosa PAO1, which has previously been linked with Salmonella response to space- and LSMMG conditions. The observations during the present study in combination with a decreased immune responsiveness of astronauts could become problematic during long-term spaceflight missions and should direct future research regarding the risk assessment and prevention of infections caused by this opportunistic pathogen. Moreover, recent work from our lab characterized the spaceflight-induced global transcriptional and proteomic responses of P. aeruginosa. Specfically, P. aeruginosa responded to spaceflight conditions through differential regulation of 167 genes (including pathogenesis-related genes) and 28 proteins, with Hfq as a global transcriptional regulator. Since Hfq was also differentially regulated in spaceflight-grown Salmonella, Hfq represents the first spaceflight-induced regulator acting across bacterial species. In addition, recent studies from our collaborative team with Staphylococcus aureus have reported molecular genetic responses to LSMMG culture that reflect those previously reported with Salmonella and Pseudomonas. This work has been submitted for publication.
Bibliography Type:	Description: (Last Updated: 07/02/2025)
Articles in Peer-reviewed Journals	Crabbé A, Pycke B, Van Houdt R, Monsieurs P, Nickerson C, Leys N, Cornelis P. "Response of Pseudomonas aeruginosa PAO1 to low shear modelled microgravity involves AlgU regulation." Environ Microbiol. 2010 Jun;12(6):1545-64. Epub 2010 Mar 5. PMID: 20236169 , Jun-2010
Articles in Peer-reviewed Journals	Crabbé A, Schurr MJ, Monsieurs P, Morici L, Schurr J, Wilson JW, Ott CM, Tsaprailis G, Pierson DL, Stefanyshyn-Piper H, Nickerson CA. "Transcriptional and proteomic responses of Pseudomonas aeruginosa PAO1 to spaceflight conditions involves Hfq regulation and reveals a role for oxygen." Appl Environ Microbiol. 2011 Feb;77(4):1221-30. Epub 2010 Dec 17. PMID: 21169425 , Feb-2011
Articles in Peer-reviewed Journals	Sarker S, Ott CM, Barrila J, Nickerson CA. "Discovery of spaceflight-regulated virulence mechanisms in Salmonella and other microbial pathogens: novel approaches to commercial vaccine development. " Gravitational and Space Biology 2010 Sep;23(2):75-8. http://gravitationalandspacebiology.org/index.php/journal/article/view/495 , Sep-2010

Articles in Peer-reviewed Journals	Jennings ME, Quick LN, Soni A, Davis RR, Crosby K, Ott CM, Nickerson CA, Wilson JW. "Characterization of the Salmonella enterica serovar Typhimurium ydcI gene, which encodes a conserved DNA binding protein required for full acid stress resistance." J Bacteriol. 2011 May;193(9):2208-17. Epub 2011 Mar 11. PMID: 21398541 , May-2011
Articles in Peer-reviewed Journals	Radtke AL, Wilson JW, Sarker S, Nickerson CA. "Analysis of interactions of Salmonella type three secretion mutants with 3-D intestinal epithelial cells." PLoS One. 2010 Dec 29;5(12):e15750. PubMed PMID: 21206750 , Dec-2010
Articles in Peer-reviewed Journals	Barrila J, Radtke AL, Crabbé A, Sarker SF, Herbst-Kralovetz MM, Ott CM, Nickerson CA. "Organotypic 3D cell culture models: using the rotating wall vessel to study host-pathogen interactions." Nat Rev Microbiol. 2010 Nov;8(11):791-801. Review. PubMed PMID: 20948552 , Nov-2010
Awards	Nickerson CA, Ott CM. "NASA Johnson Space Center Director's Innovation Group Achievement Award, May 2010." May-2010
Significant Media Coverage	Harrington JD, Caspermeyer J. "'Spacebound Bacteria Inspire Earthbound Remedies.' NASA press release about PI's research findings." NASA Media Advisory 11-080, March 21, 2011. http://www.nasa.gov/home/hqnews/2011/mar/HQ_11-080_Bacteria_Research.html , Mar-2011
Significant Media Coverage	Science Daily staff. "'Out of This World: New Study Investigates Infection of Human Cells in Space.' News article about PI's research and flight experiment. " ScienceDaily, April 4. 2010. http://www.sciencedaily.com/releases/2010/04/100402154914.htm , Apr-2010